# 6.2 Radicals and Rational Exponents

Name _____ Period ____ Date _____

Algebra I 6.2 · Student sheet

**I can…**

- find nth roots.
- rewrite expressions between radical form and rational exponent form.
- evaluate expressions with rational exponents.

## Part 1 · Exponent detective

5^(1/2) · 5^(1/2) = 5¹ = 5. So 5^(1/2) is the number that times itself makes 5. That number is _____

8^(1/3) · 8^(1/3) · 8^(1/3) = 8¹ = 8. So 8^(1/3) = _____

| Expression | 9^(1/2) | 16^(1/2) | 8^(1/3) | 27^(1/3) | 16^(1/4) |
|---|---|---|---|---|---|
| Value | | | | | |

Rule: a^(1/n) means _____

## Part 2 · Notes: fill in the blanks

$\sqrt[n]{a}$ is a number whose _____ power is a. The n is called the _____.

If n is odd, a has _____ real nth root. If n is even and a > 0, a has _____ real nth roots.

a^(1/n) = _____ a^(m/n) = _____

In a^(m/n), the denominator tells the _____ and the numerator tells the _____.

## Part 3 · Examples (work them with the class)

### Example 1

Rewrite each expression in radical form.

a. $10^{\left(\frac{1}{2}\right)}$ b. $11^{\left(\frac{1}{5}\right)}$

### Example 2

Find the indicated real nth root(s) of a.

a. n = 3, a = −27 b. n = 4, a = 16

### Example 3

Evaluate each expression.

a. $\sqrt[3]{(-64)}$ b. $-\sqrt[4]{16}$

### Example 4

Evaluate each expression.

a. $16^{(\frac{3}{4})}$ b. $27^{(\frac{2}{3})}$ c. $32^{(-\frac{2}{5})}$

### Example 5

A cube has a volume of 1331 cubic feet. Find the length of each side.

## Part 4 · Partner practice

### Partner 1

Evaluate $64^{(\frac{2}{3})}$.

### Partner 2

Evaluate $81^{(-\frac{1}{4})}$.